

Acme E-Scooter X1

Service Manual

1. Product Overview

The Acme E-Scooter X1 is a 36V lithium-ion electric scooter with a 350W brushless hub motor, rated for a 25 km range. This manual covers the electrical system, braking, and routine service. Always power off and remove the key before opening any panel.

2. Components

- Front panel — houses the fuse block, headlight wiring, and display ribbon connector.
- Fuse F3 (10A), beneath front panel, Figure 4.2 — protects the headlight and display circuit.
- Main controller (located in the deck) — drives the hub motor and reads the throttle Hall sensor.
- 36V 7.8Ah battery pack — under the footboard, secured by four M4 screws.
- Throttle assembly — Hall-effect twist grip wired to the controller via a 3-pin JST connector.
- Rear disc brake with electronic cutoff switch that disables the motor when the lever is pulled.

3. Common Issues, Causes & Fixes

3.1 Scooter powers on but headlight and display are dead.

Possible causes:

- Blown Fuse F3 (10A) beneath the front panel (Figure 4.2).
- Loose display ribbon connector at the front panel.

Fix: Open the front panel, inspect Fuse F3. If the filament is broken or glass is darkened, replace with a 10A automotive blade fuse. Reseat the display ribbon connector. Do not bypass the fuse with foil.

3.2 Motor does not spin when throttle is applied, no error code.

Possible causes:

- Brake cutoff switch stuck engaged (motor inhibited).
- Throttle Hall connector unplugged at the controller.

Fix: Check that the brake lever returns fully; a stuck cutoff switch keeps the motor disabled. Reseat the 3-pin throttle connector at the controller. Verify 4.2V on the throttle signal wire at rest.

3.3 Range dropped sharply and scooter shuts off under load.

Possible causes:

- Aged or unbalanced 36V battery pack.
- Loose battery main connector causing voltage sag.

Fix: Charge to full, then measure resting pack voltage (should be ~42V full). If it sags below 30V under throttle, the pack needs balancing or replacement. Retorque the main battery connector.

4. Maintenance Schedule

Task	Interval	Notes
Replace / inspect front Fuse F3	Every 6 months	10A blade fuse, beneath front panel (Fig 4.2).
Brake pad inspection and adjustment	Every 3 months	Replace pads below 2mm.
Tire pressure check	Monthly	Maintain 45 PSI front and rear.
Battery balance charge	Every 6 months	Leave on charger 2h past full.